Olympic 5000m race prediction: Tweet sentiment vs. past performances

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Introduction

- **Focus:** Predicting the men's 5000m sprint results at the 2024 Paris Olympics.
- **Context:** Sports predictions impact betting, sponsorship, and training with most research centered on team sports.
- **Past Findings:** Neural networks showed 77 percent of accuracy in horse race predictions [3], while football predictions had no clear advantage over betting odds [5]. A sentiment analysis of Tweets about Premier League matches led to a higher payout but lower accuracy compared to betting odds favorites. [4]

Research question:

Can we replicate the past findings for an individual sport?

Our proposal

To answer the research question we are proposing two separate approaches to compare their predictive performance. The methodology is as follows:

Sentiment Analysis
Sentiment analysis of recent and most popular tweets that mention the athletes' names before the race.

AI models
Use several AI models incorporating freely accessible and simulated race features to predict the rank of the athletes.

Evaluation metrics

- **Sentiment analysis:** Correlation
- **AI models:** Mean Absolute Error, Mean Square Error, Root Mean Square Error

Dataset overview

For the two proposed models we use two separate datasets for the 22 starters in the race final.

Tweets
A total of 625 Tweets were collected. The data includes the athlete used as the keyword to obtain the Tweet, the Tweet content, the username of the person who posted it, and the posting date, combined with the athlete's actual race time and rank.

Performances related features
100 first-ranked athletes in the world for the 5000m sprint race.
Variables : Athlete, Nationality, Venue of the race, World Rank, Date, Score [1].
Added Simulated varibales: Crowed Cheering effect, Environmental Adaptation.

We evaluated the models on the actual race results.

Data recollection - Web scraping and Twikit package

Obtaining the Tweets presented several challenges:

- After Elon Musk acquired X (formerly Twitter), free API access was eliminated in February this year.
- Due to ongoing competitions over the summer, the starter list was announced only shortly before the start of the Olympics.
- X actively works to prevent large-scale web scraping of Tweets, banning accounts that exhibit suspicious activity.
- Web scraping large volumes of Tweets is computationally intensive.

Twikit, a Python package for web scraping Twitter data, allowed us to access Tweets without needing API keys. To collect Tweets about the 22 athletes, we applied filters to capture recent and popular Tweets, obtaining approximately 30 Tweets per athlete.

Sentiment analysis

For sentiment analysis, we utilized the *Hugging Face* package. After exploring various models, we present the most successful one:

nlptown/bert-base-multilingual-uncased-sentiment

The 'nlptown/bert-base-multilingual-uncased-sentiment' model belongs to the family of **BERT** (Bidirectional Encoder Representations from Transformers) based models, a natural language processing (**NLP**) framework developed by Google. BERT models are pretrained on large text corpora like Wikipedia. This particular model is fine-tuned for sentiment analysis, supports six languages, and **categorizes** sentiments into **five classes**: very negative, negative, neutral, positive, and very positive.

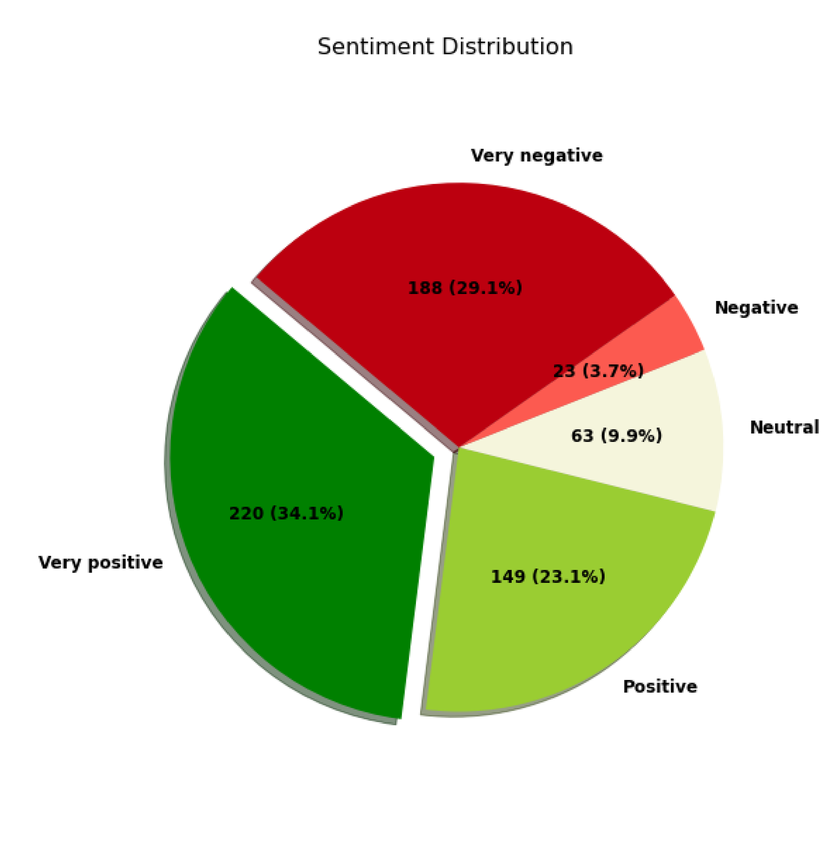

Figure 1. Sentiment classification of the Tweets

Past Performances and Simulated Features Models

To predict 5000m sprint **scores** (a combination of time and placement in a race [1]), we explored various models using past performances and simulated features. We then compared three different metrics (Table 1) and analyzed the residual plots to select the best model.

1. Linear Regression & Polynomial Regression

- Used to model the relationship between time and features.
- Polynomial Regression extends Linear Regression by fitting a polynomial curve.

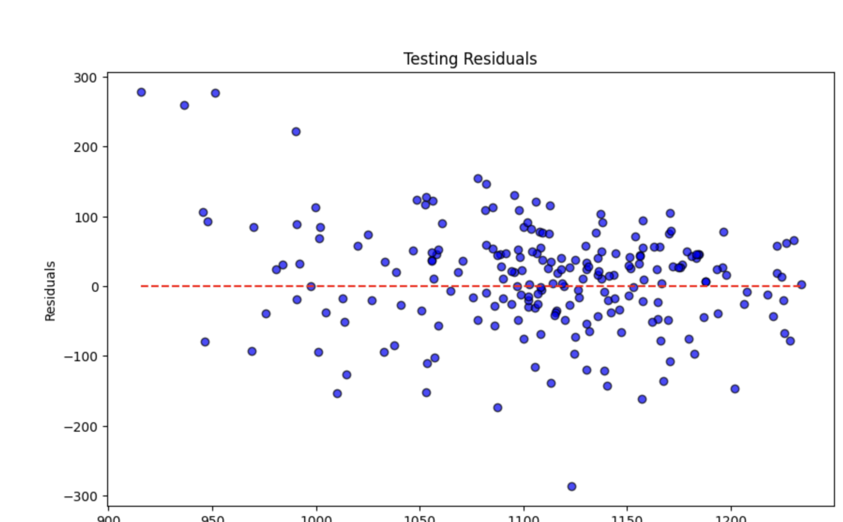

Figure 2. Residual plots of the Linear Regression Testing Set

2. Neural Network

- Employed a feedforward neural network.
- Captures complex, non-linear relationships.

3. Random Forest

- An ensemble method using multiple decision trees.
- Handles non-linearities and interactions well.
- **Best performing model.**

Table metrics

	MAE	MSE	RMSE
Linear Regression	57.87	5798.98	76.15
Neural Networks	54.18	5884.81	76.71
Random Forest	45.31	3829.40	61.88

Table 1. Comparison between metrics and models

Findings/Conclusion

Sentiment analysis:

To evaluate if sentiment could predict race results, we analyzed its correlation with athletes' race times and ranks.

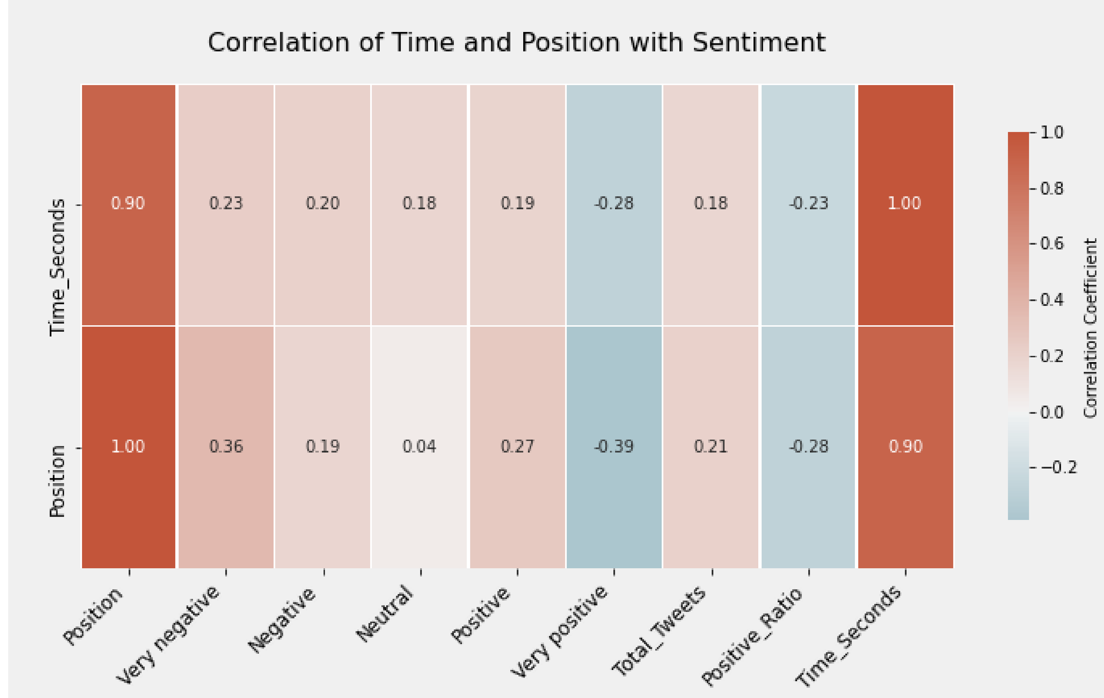

Figure 3. Correlation between Sentiment and race results

We found a moderate positive correlation between very negative Tweets and athlete rank, indicating that more negative Tweets are linked to higher ranks. Conversely, there's a moderate negative correlation with very positive Tweets. Correlations with race time were weak, with the strongest ties again in extreme sentiments.

Given these small to medium correlations, we concluded sentiment alone is not a reliable predictor of rank or time. This might be caused by the data limitations, which could improve with more Tweets accessed via the API.

AI models:

The Random Forest outperformed other models, showing the lowest prediction error and better capturing the intricate patterns in the data.

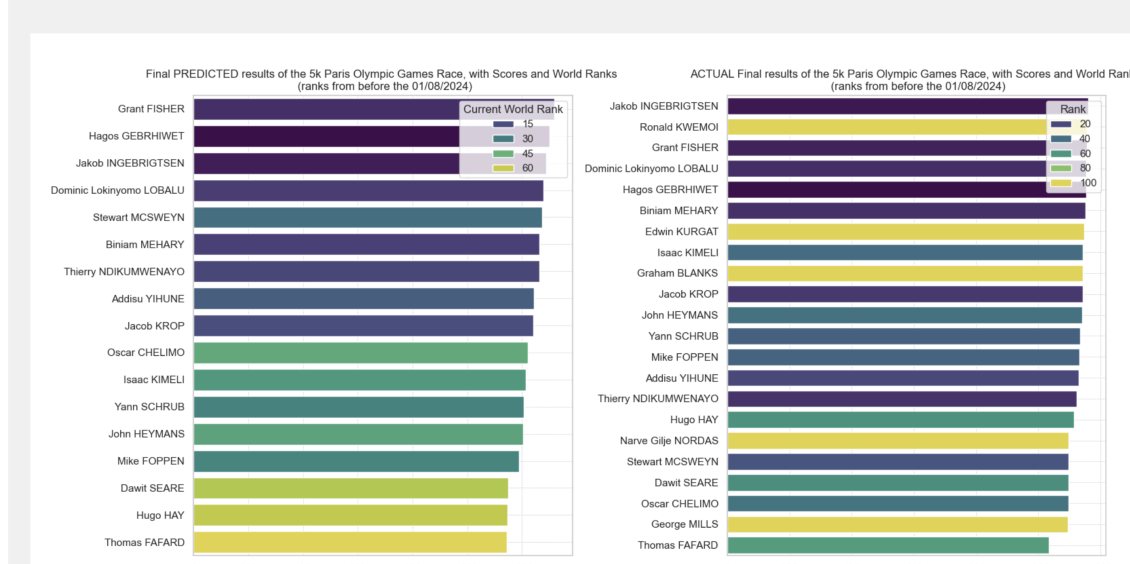

Figure 4. Comparison Predicted and Actual Results

Conclusion

Although our models perform reasonably well, they do not provide complete certainty in our predictions. To enhance accuracy, we recommend the following improvements:

- Integrating sentiment analysis with AI models
- Consulting with field experts to gain deeper insights into the variables
- Acquiring access to a larger dataset
- Further refining the AI models through hyperparameter tuning, adjusting the number of layers, and optimizing the learning rate

References

[1] World Athletics.

[2] University College Dublin.

[3] Elnaz Davoodi and Ali Reza Khanteymoori. Horse racing prediction using artificial neural networks. *Recent Advances in Neural Networks, Fuzzy Systems & Evolutionary Computing*, 2010:155–160, 2010.

[4] Robert P. Schumaker, A. Tomasz Jarmoszko, and Chester S. Labedz. Predicting wins and spread in the premier league using a sentiment analysis of twitter. *Decision Support Systems*, 88:76–84, 2016.

[5] Niek Tax and Yme Joustra. Predicting the dutch football competition using public data: A machine learning approach. *Transactions on knowledge and data engineering*, 10(10):1–13, 2015.